

Compared to What? Know the Language First

Module 2 · Day 1, Session 1: Five Words for Reading Any Result

3ie

0 · Where Module 1 left us

The word Module 1 was built on

From Module 1: what is a **counterfactual**?

What would have happened to the same people, over the same period, without the programme. We never observe it. Every evaluation method estimates it with a **comparison**, and a result can only be read once you know what that comparison was.

Today we attach five words to that idea, using one programme and its data from start to finish.

1 · What are we comparing?

The traffic camera programme

In 2020 a roads authority installed automated speed cameras in twelve sectors of its road network. The other twelve sectors were due to get them two years later.

Before extending the programme, someone has to decide whether the cameras worked.

Programme	Automated speed cameras
Roads	1,200 road segments in 24 sectors
Measured	2019, before; 2021, after
Decision	Extend to the other twelve sectors?

Illustrative case, with invented figures. This is not a real programme; the numbers are generated for this training.

The headline

-32%

Fatal collisions on camera roads, 2019 to 2021

Before **233** After **159**

[MENTI PLACEHOLDER] On this number alone, would you extend the cameras to the other twelve sectors?

Go to **menti.com**, code **1234 5678**, or scan the code.



Illustrative case, with invented figures.

What if traffic had changed?

Change in driving on camera
roads (%)



```
fatal <- tapply(cam$fatal_collisions, cam$year, sum)
km      <- sum(cam$vehicles_km_millions[cam$year == 2019]) * c(1, 1 + traffic / 100)
round(data.frame(fatal_change_pct = 100 * (fatal[2] / fatal[1] - 1),
                 risk_per_km_change_pct = 100 * ((fatal[2] / km[2]) / (fatal[1] / km[1]) - 1),
                 row.names = NULL))
```

fatal_change_pct	risk_per_km_change_pct
-32	-32

The count fell 32%. Slide driving down to -32%. **What happens to the risk per kilometre driven?**

It stops moving. If people drove a third less, a third fewer deaths is what you would expect with or without cameras. The headline alone cannot tell the two apart.

What actually happened to traffic

R Code

↺ Start Over

▶ Run Code

```

1 fatal <- tapply(cam$fatal_collisions, cam$year, sum)
2 km     <- tapply(cam$vehicles_km_millions, cam$year, sum)
3 round(data.frame(fatal_change_pct = 100 * (fatal[2] / fatal[1] - 1),
4                 driving_change_pct = 100 * (km[2] / km[1] - 1),
5                 risk_per_km_change_pct = 100 * ((fatal[2] / km[2]) / (fatal[1] / km[1]) - 1),
6                 row.names = NULL), 1)

```

fatal_change_pct	driving_change_pct	risk_per_km_change_pct
-31.8	4.4	-34.7

Driving on camera roads went **up** by 4.4%. So the risk per kilometre fell by 35%, a little more than the headline said.

The comparison moved the answer, this time in the programme's favour. What other comparison would you want before crediting the cameras?

“Compared to what?”

Every result is a comparison. A before-and-after number compares the programme with **its own past**. What else changed over the same period (traffic, vehicles, a national campaign) is inside that number, and it can push the answer either way.

Think of a figure you reported, or were sent, this year. **What was it compared to?** If nobody said, what would you have picked?

2 · Three questions for any number

One idea, three questions

The five words that follow each answer one of three questions you can ask of any number an evaluator hands you.

Compared to what?	Comparison group, counterfactual
How big, in units that matter?	Mean · treatment effect · coefficient
How sure are we?	p-value · confidence interval

We come back to this list at the end of the session, and on every day of the week.

3 · The five words

Word 1 · Mean: ten roads, by hand

Road	Type	Injury collisions
SEC08-013	urban arterial	5
SEC10-021	suburban arterial	6
SEC10-024	urban arterial	7
SEC12-020	ring road	7
SEC04-033	urban arterial	8
SEC02-023	ring road	12
SEC10-047	school corridor	13
SEC08-002	suburban arterial	16
SEC06-011	suburban arterial	19
SEC05-018	ring road	42

Ten camera roads in 2019. With pen and paper: **what is the average number of injury collisions per road?** Then: is a typical road close to it?

The mean is **13.5**. 7 of the ten roads are below it. One ring road with 42 collisions pulls the mean up; the middle road (the **median**) has 10.

Illustrative case, with invented figures.

Word 1 · What the mean hides

- All ten roads
- Without the busiest road

Remove one road. How far does the mean move, and how far does the median move?

R Code

Start Over

Run Code

```
1 roads <- ten$injury_collisions
2 if (keep == "Without the busiest road") roads <- roads[roads < max(roads)]
3 dots(roads)
```



The mean drops from 13.5 to 10.3; the median from 10 to 8. One road moved the mean further than the median. When an evaluator reports an average, ask what the spread looks like and whether a few cases drive it.

Word 2 · Treatment effect

Average speed (km/h)	2019	2021	Change
Camera roads	73.17	69.80	-3.37
Roads without cameras	67.07	67.05	-0.02
Difference	6.10	2.75	-3.35

Three numbers describe this result: **-3.37**, **2.75** and **6.10**. Work out what each one is, then decide which is the effect of the cameras on speed.

6.10 is the head start: camera roads were faster before anything happened, and **2.75** still carries it. The **treatment effect** is the change on camera roads minus the change on the others: **-3.35** km/h. It is close to the camera roads' own change only because the other roads barely moved.

Illustrative case, with invented figures.

Word 2 · The same number from 1,200 roads

R Code

↺ Start Over

▶ Run Code

```
1 cams <- mean(wide$change[wide$treated == 1])
2 others <- mean(wide$change[wide$treated == 0])
3 round(data.frame(camera_roads = cams, other_roads = others, effect = cams - others), 2)
```

camera_roads	other_roads	effect
-3.37	-0.02	-3.36

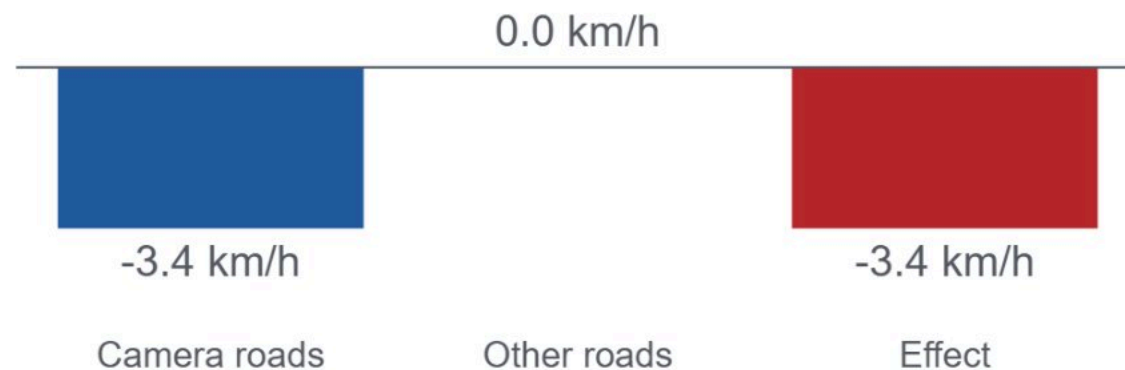
`wide` holds one row per road: its speed in 2019, in 2021, and the change. The three lines are the arithmetic you just did, on every road.

Word 2 · The effect depends on the comparison

Speed change on roads without cameras (km/h)



```
cams <- mean(wide$change[wide$treated == 1])
bars(c("Camera roads" = cams, "Other roads" = other, "Effect" = cams - other))
```



Camera roads are fixed at -3.4. Move the other roads. **Which value makes the effect zero, and what would that mean in plain words?**

Word 3 · Coefficient

R Code

↺ Start Over

▶ Run Code

```
1 fit <- lm(change ~ treated, data = wide)
2 round(data.frame(coefficient = coef(fit)), 2)
```

	coefficient
(Intercept)	-0.02
treated	-3.36

`treated` is 1 for camera roads, 0 for the rest. A **coefficient** says how much the outcome moves per one unit of the row. Here one unit is “has cameras”.

Which of the two numbers is the effect? What is the other one?

`treated` is **-3.36**, the effect you worked out by hand. `(Intercept)` is the average change on roads without cameras. It is not an effect.

Word 3 · The other rows of a table

R Code

↺ Start Over

▶ Run Code

```
1 fit2 <- lm_robust(change ~ treated + school_within_500m + lanes, data = wide)
2 round(as.data.frame(summary(fit2)$coefficients[, 1:4]), 3)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.096	0.253	0.380	0.704
treated	-3.344	0.136	-24.636	0.000
school_within_500m	-0.371	0.163	-2.280	0.023
lanes	-0.009	0.061	-0.145	0.885

Std. Error is how much the estimate would move from one sample of roads to another. Small relative to the estimate means a steady estimate.

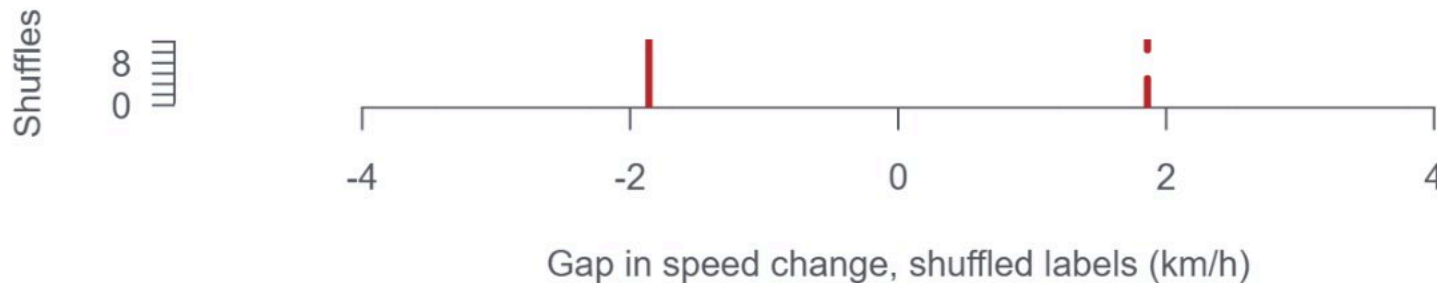
The other rows adjust the comparison. `school_within_500m` says roads near schools slowed a little more. That is not a recommendation to build schools next to roads.

Word 4 · p-value: shuffle the labels

Times we shuffle the camera labels



```
small <- study(20)      # a study of 20 roads: 10 with cameras, 10 without
shuffle_plot(small, k)
```



Measured gap: -1.9 km/h. Shuffles: 0. As far from zero by chance: 0 -> $p = ?$

If the cameras did nothing, the labels would be arbitrary, so we shuffle them. **How often does chance alone give a gap as large as the red line?** Guess first.

That share is the **p-value**: here about 0.17, one shuffle in six.

Word 4 · Significant is not the same as big

Suppose cameras cut speed by only (km/h)

3.4



`significance_demo(cut_by)`

All 1,200 roads. Effect -3.4 km/h, 95% interval -3.67 to -3.13, $p = <0.001$



Slide the effect down to 0.3 km/h, a change no driver would notice. **Is it still statistically significant?**

Yes, p is about 0.03. With 1,200 roads even a trivial change passes the test. Whether 0.3 km/h is worth paying for is a separate question, and the p -value cannot answer it.

Word 4 · The warning cuts both ways

Roads in the study



```
set.seed(3)
p <- replicate(200, t.test(change ~ treated, data = study(roads_n, seed = sample(1e6, 1)))$p.value)
data.frame(roads_in_study = roads_n, share_of_studies_that_detect_it = round(mean(p < 0.05), 2))
```

roads_in_study	share_of_studies_that_detect_it
20	0.8

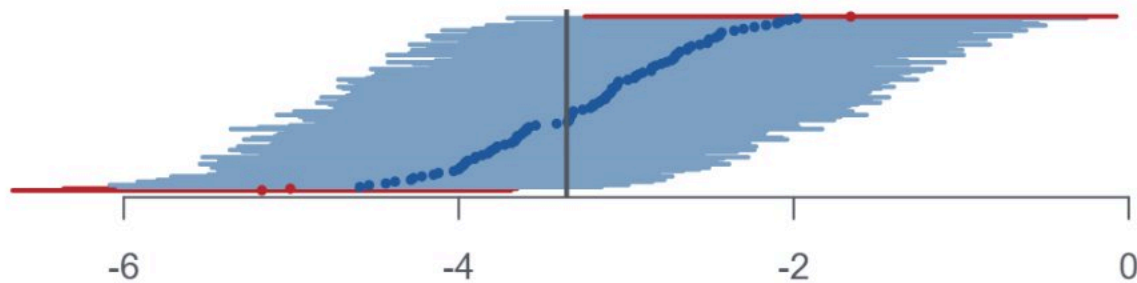
The real effect is -3.4 km/h on every run. What changes is the size of the study. At 20 roads, about one study in five misses it, like the $p = 0.17$ study two slides ago.

An evaluator reports “no significant effect”. Ask two things: **how many roads (or people) were in the study, and what is the smallest effect it could have detected?**

Word 5 · Confidence interval

Run another 100 studies

```
many_studies(n = 40, runs = 100, seed = run + 1)
```



Effect on average speed (km/h), one line per study

100 studies of 40 roads. 97 intervals contain the all-roads value (-3.36); 3 miss it.

Each line is one study of 40 roads and its **95% confidence interval**, the range of values consistent with its data. **Out of 100, how many will miss the grey line?** Guess before the next press.

Word 5 · Which one would you act on?

Suppose three evaluations of the cameras each reported **1.8 fewer injury collisions per road**, and the authority's bar is **2.0**. Only the intervals differ.

- A** 1.8 fewer, 95% interval 0.4 to 3.2
- B** 1.8 fewer, 95% interval 1.6 to 2.0
- C** 1.8 fewer, 95% interval -0.9 to 4.5

Which would you act on? What is your answer for each?

B is precise and below the bar: confidently not enough. A is too uncertain to call, and C more so. All three have the same estimate, so the intervals make the difference.

4 • Put AI to work

AI prompt cards

Each group takes one card. Paste its prompt into your AI tool, read the answer together, and mark anything it gets wrong or overstates.

- Card 1** "Explain a 95% confidence interval, in one paragraph, for someone who is not a statistician."
- Card 2** "Explain what a p-value means."
- Card 3** "Fatal collisions on our camera roads fell 32%. Is that a good result?"
- Card 4** "The coefficient on school_within_500m is -0.37. What should I do about it?"

Card 1. A response like this is possible:

"A 95% confidence interval is the range in which the true value is likely to fall. **There is a 95% probability that the true effect lies inside this interval**, so if the interval is -3.6 to -3.1 km/h the true effect is probably in that range. A narrower interval means a more precise estimate."

Debrief

Card 1. The flagged sentence is the common error. "95%" describes the method: over many studies, 95 of 100 intervals catch the true value, as the chart of 100 studies showed. It is not a probability for this one interval.

Card 3. Did the AI ask what the 32% was compared with, or how much driving changed? The prompt gave it neither.

Card 4. Look for advice about schools. The row adjusts the comparison; it is not a policy lever.

Two habits: ask the AI to **ask you questions before it answers**, and put its answer through a second chat to check it. Also note what it explained well.

5 • Triage the claims

Triage the claims

1. "Fatal collisions on camera roads fell 32% over two years."
2. "The effect of cameras on speed is -3.4 km/h, 95% confidence interval -3.6 to -3.1."
3. "The fall in speed is statistically significant ($p < 0.001$), so every road should get a camera."
4. "Speed fell 3.4 km/h on camera roads and did not change on roads without them."
5. "The average road in our sample had 13.5 injury collisions."
6. "A study of 20 roads found no significant change in speed."
7. "The coefficient on school_within_500m is -0.37, so schools near roads slow traffic."
8. "Speed fell 3.4 km/h more than on roads without cameras; the target was at least 3 km/h."

In groups: for each excerpt, **which of the three questions does it leave unanswered, and would you act, not act, or ask first?** Six minutes. Then: which one were you most tempted to act on?

6 · Take this back to your desk

The three questions

From memory, before you look: what are the three questions you can ask of any number?

- 1 Compared to what?
- 2 How big is it, in units that matter to the decision?
- 3 How sure are we?

Next session opens the full camera data file and asks these three questions of the collision numbers.